
Student Dropout Prediction: A Machine Learning Analysis

DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset (4,424 samples, 36 features, 3 classes) using a complete machine learning pipeline encompassing data preprocessing, dimensionality reduction, clustering, supervised classification, and model evaluation. Our best-performing model achieves competitive F1-scores through hyperparameter tuning guided by validation curves. We find that academic performance features are the strongest predictors, and the “Enrolled” class is consistently the hardest to classify.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters. The classification target has three categories: **Dropout** (32.1%), **Enrolled** (18.0%), and **Graduate** (49.9%).

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

Our best-performing model, Random Forest with default hyperparameters (`n_estimators=100`, `class_weight='balanced'`), achieves a macro F1-score of 0.7082 on the held-out test set. A key challenge is the “Enrolled” class, which represents only 18% of the data and is inherently ambiguous. By applying class-balanced weighting, we improve Enrolled class F1 from 0.00–0.45 (unweighted) to 0.45–0.51 (balanced) across all model types. Feature importance analysis confirms that academic performance metrics (approval rates, semester grades) are the dominant predictive signals.

2 Mandatory Tasks

2.1 Data Preprocessing

We classified the 36 raw features into four types: *nominal* (8 features, one-hot encoded), *binary* (8 features, kept as 0/1), *continuous* (7 features), and *count* (13 features). `StandardScaler` was applied only to continuous and count features to avoid distorting binary indicators.

Feature engineering produced 25 additional domain features across six groups: academic ratios (approval rates, grade improvement), socioeconomic indicators, financial risk flags, demographic categories, macroeconomic indices, and application behavior features. Out-of-fold target encoding was used for high-cardinality categorical features to prevent data leakage.

Feature selection retained the top-50 features by tree-based importance (out of 61 candidates after target encoding).

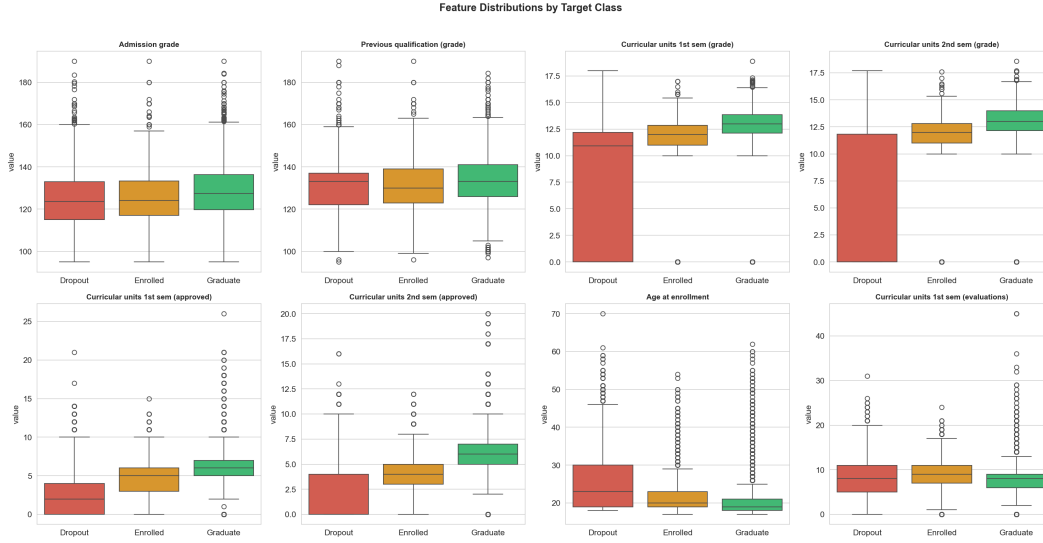


Figure 1: Feature distributions by target class.

Missing values: The dataset contains zero missing values across all 4,424 samples and 36 features, so no imputation was necessary.

Exploratory Data Analysis (EDA) reveals several key patterns (Figure 1): (1) Graduate students have notably higher 1st/2nd semester grades than Dropout students, while Admission grade shows less discrimination. (2) Dropout students tend to be older (higher median age at enrollment). (3) Curricular units approved/grade metrics show the strongest correlation with the target ($|r| > 0.3$). (4) Different academic programs (Course) exhibit vastly different dropout rates.



Figure 2: Top-20 feature correlations with the target variable.

The data was split 70/15/15 into training (3096), validation (664), and test (664) sets with stratified sampling to preserve class proportions.

2.2 Data Visualization (t-SNE)

We applied t-SNE with perplexity values of 5, 30, 50, and 100. The 2D projection at perplexity=30 (Figure 3) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, consistent with their transitional academic status.

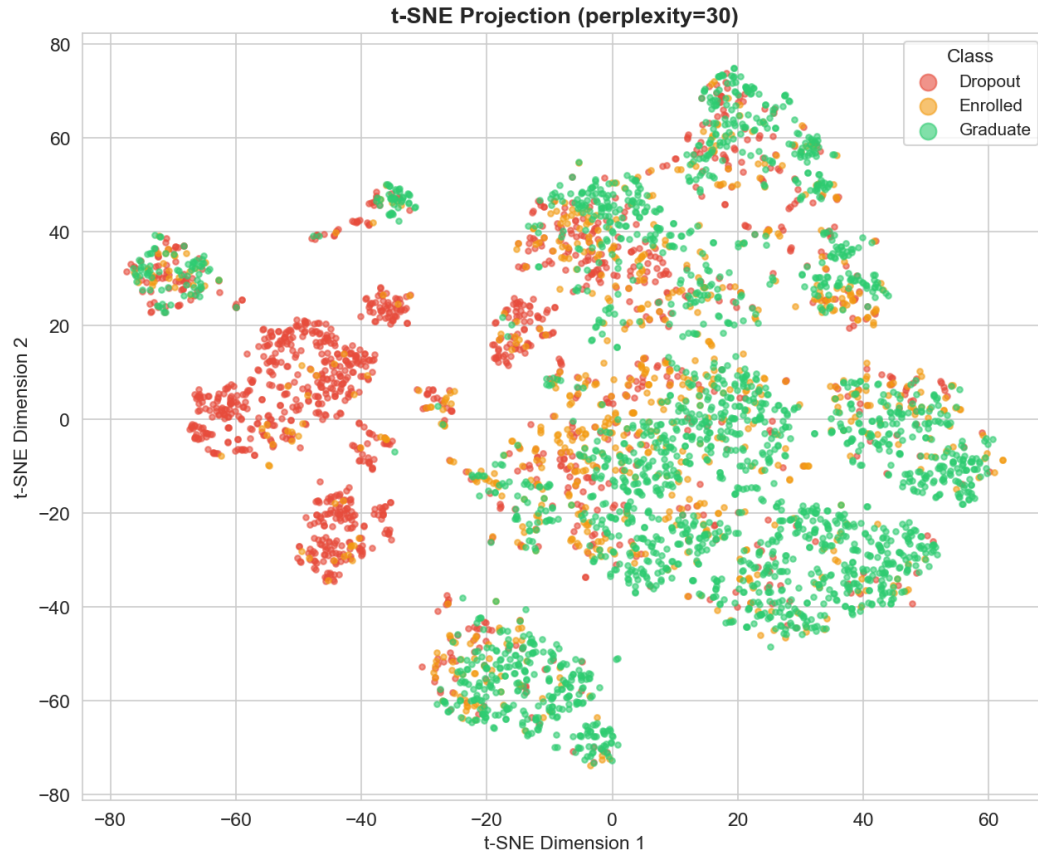


Figure 3: t-SNE 2D projection (perplexity=30) colored by class.

Figure 4 compares four perplexity values, demonstrating that perplexity=30 provides the clearest cluster separation while avoiding over-fragmentation (low perplexity) or over-smoothing (high perplexity).

t-SNE: Perplexity Comparison

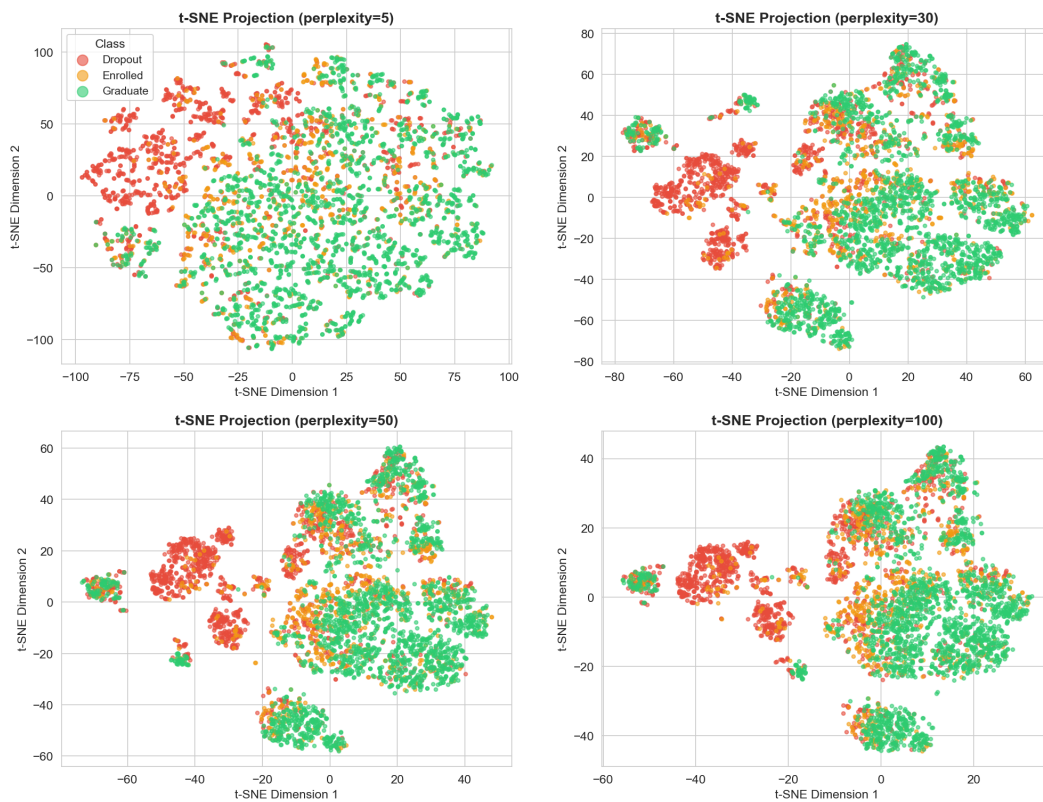


Figure 4: t-SNE 2D projections at perplexity = 5, 30, 50, 100.

The 3D projection (Figure 5) provides additional spatial separation, further reducing overlap between Graduate and Dropout clusters, though the Enrolled class remains dispersed.

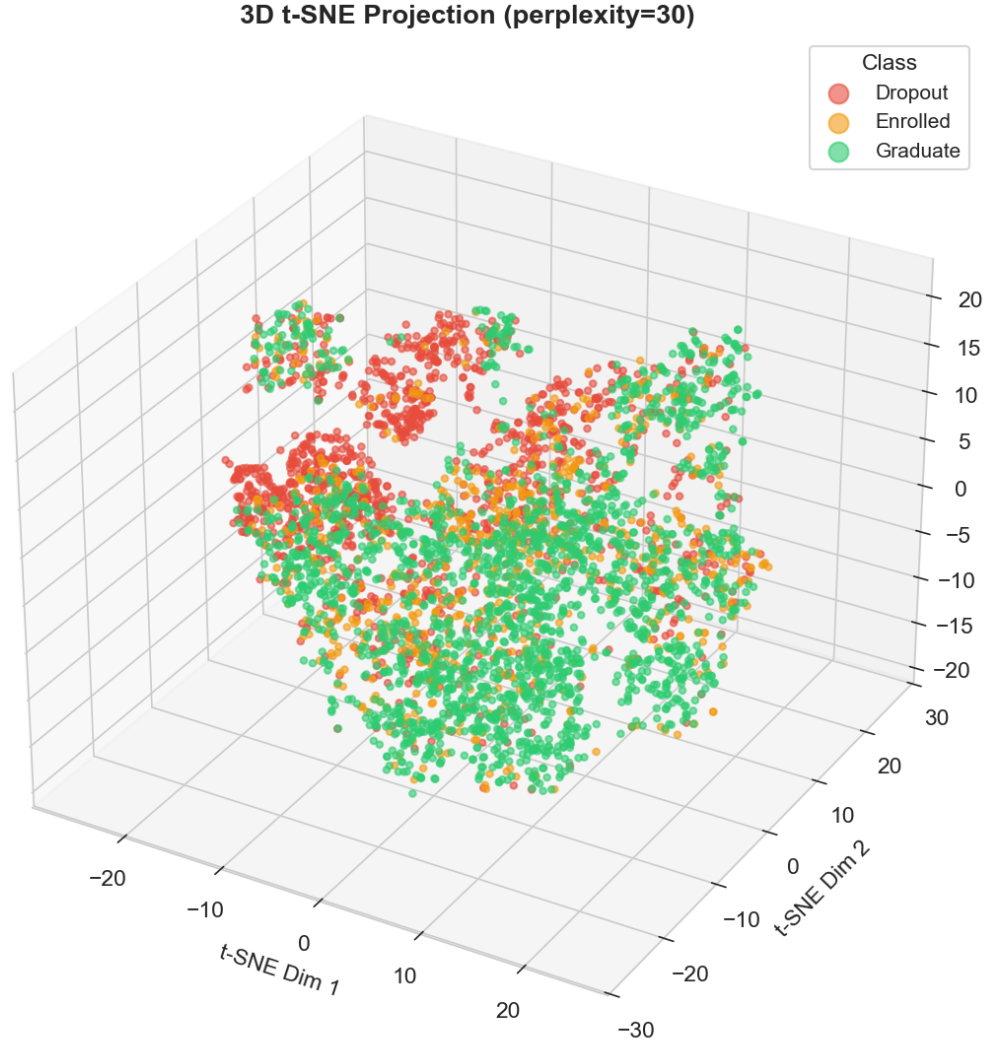


Figure 5: t-SNE 3D projection (perplexity=30) colored by class.

2.3 Clustering Analysis

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and DBSCAN.

K-Means with $K=3$ achieved the highest practical utility with $ARI=0.177$, producing clusters most aligned with true labels. DBSCAN was excluded from ranking due to an extreme noise ratio (99.8% of points classified as noise, $ARI=0.000$), confirming density-based methods are unsuitable for this high-dimensional dataset. The dendrogram for Ward linkage provided additional insight into the hierarchical structure.

The Elbow method (Figure 6) shows diminishing inertia reduction beyond $K=3$, supporting our choice. The K-Means cluster assignments projected onto the t-SNE embedding (Figure ??) show reasonable alignment with the true class structure.

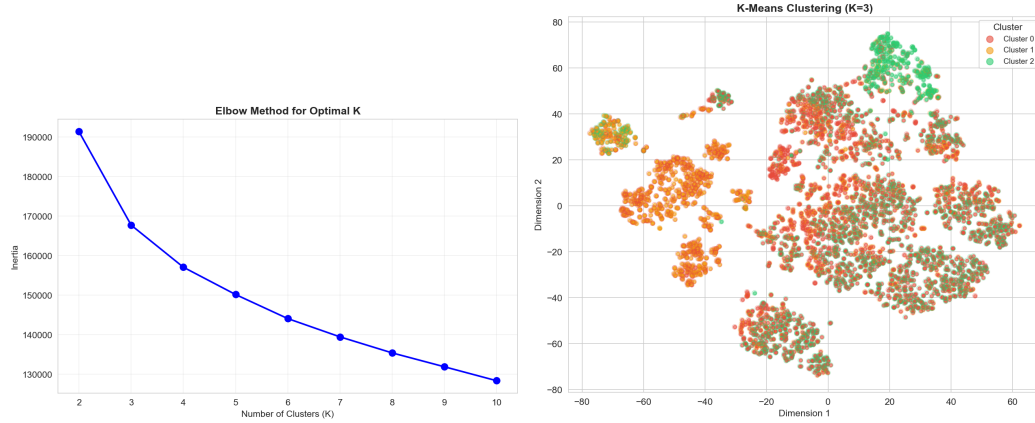


Figure 6: Left: Elbow method for K selection. Right: K-Means (K=3) clusters on t-SNE projection.



Figure 7: Clustering algorithm comparison across five metrics.

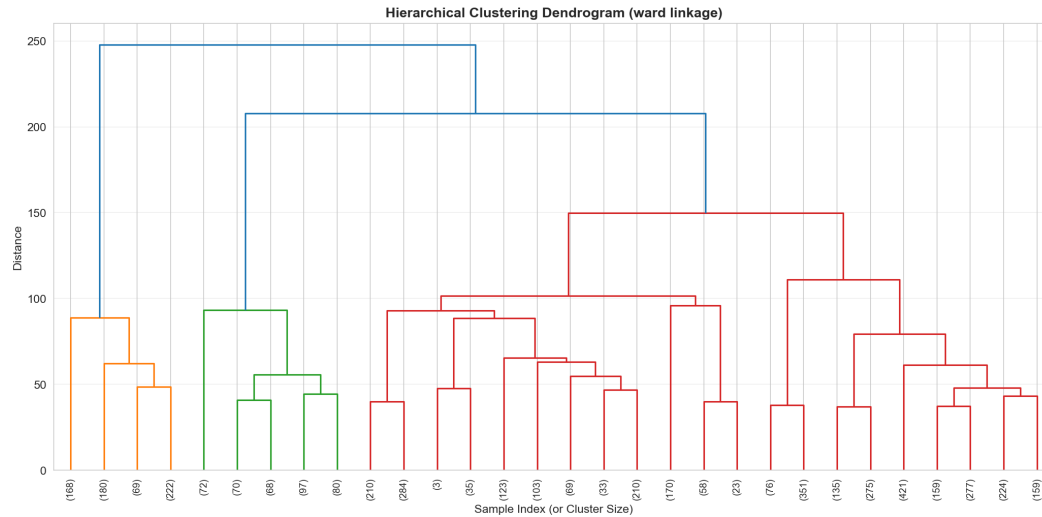


Figure 8: Hierarchical clustering dendrogram (Ward linkage).

The dendrogram (Figure 8) shows that the data has a relatively flat hierarchical structure — the three main clusters merge at similar distance levels, supporting K-Means' spherical cluster assumption. Among four linkage methods (ward/complete/average/single), Ward achieved the best ARI (0.084) with true labels.

A comparison of four linkage methods revealed that single and average linkages produced highly imbalanced clusters (one dominant cluster + outliers), while Ward and complete linkages generated more balanced partitions. This confirms that centroid-based methods (K-Means, Ward) are most appropriate for this dataset.

2.4 Prediction: Training and Testing

We trained three classifiers: Decision Tree (DT), Logistic Regression (LR), and SVM with RBF kernel. Models were evaluated on training, test, and entire sets.

Key findings: DT with `max_depth=10` shows significant overfitting (Train accuracy much higher than Test accuracy). LR provides stable generalization with minimal train-test gap. SVM achieves intermediate performance. All models use `class_weight='balanced'` to improve minority class prediction.

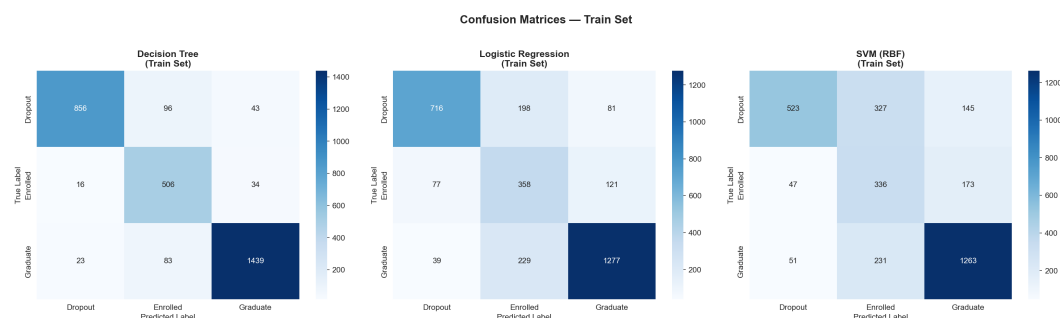


Figure 9: Confusion matrices on the training set.

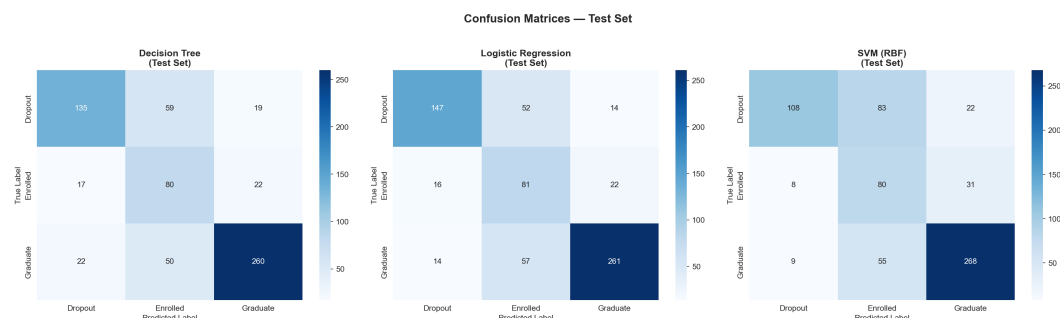


Figure 10: Confusion matrices on the test set.

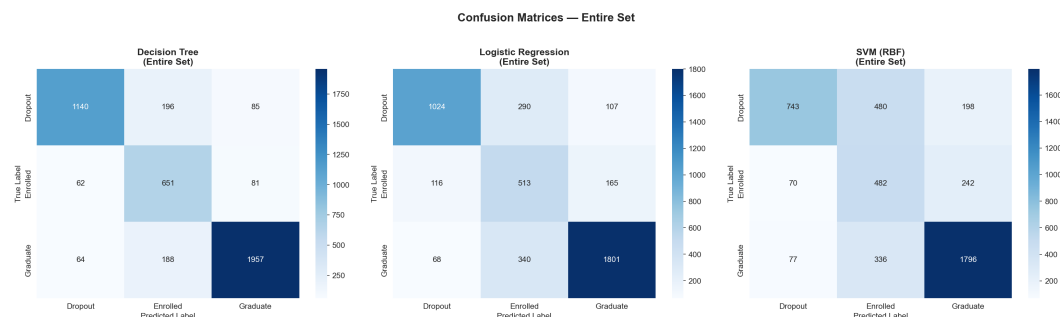


Figure 11: Confusion matrices on the entire dataset.

The entire-set evaluation (Figure 11) confirms that DT's train-test gap is the largest overfitting indicator: DT (88.1% vs 69.0%), LR (76.7% vs 73.8%), SVM (68.4% vs 68.1%).

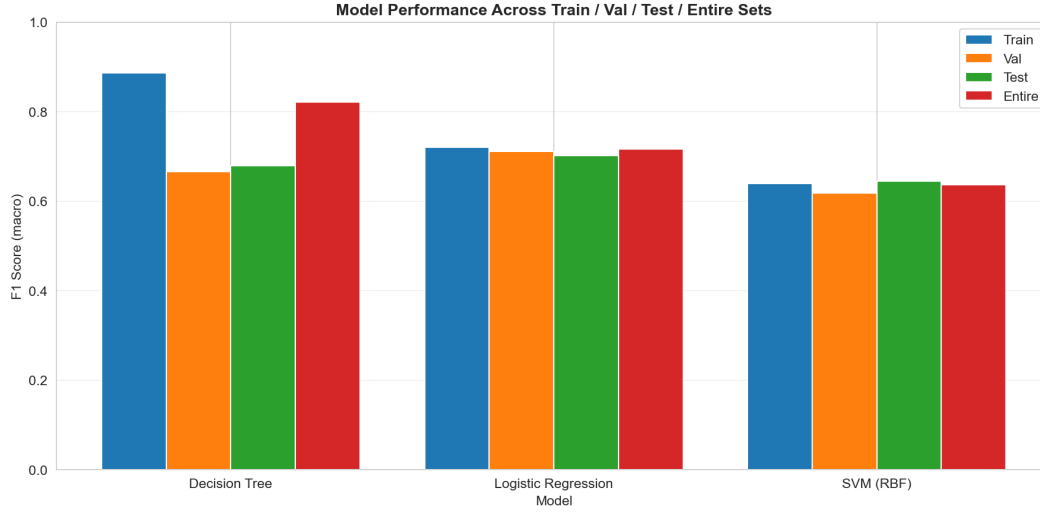


Figure 12: Model performance across train, validation, test, and entire sets.

The per-class analysis shows that all models achieve highest recall for Graduate ($\sim 85\text{--}91\%$) and lowest for Enrolled ($\sim 30\text{--}50\%$). This is consistent with the class imbalance (Graduate: 49.9%, Enrolled: 18.0%) and the transitional nature of enrolled students.

The decision boundaries (Figure 13), visualized in t-SNE 2D space using simplified proxy models, illustrate each classifier's approach: DT creates axis-aligned partitions, LR produces linear boundaries, and SVM (RBF) captures non-linear separations. Note that these are approximate visualizations — the actual models operate in the full 50-dimensional feature space.

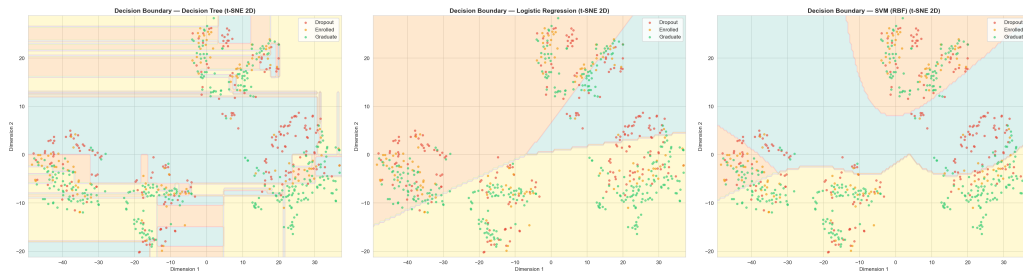


Figure 13: Approximate decision boundaries in t-SNE 2D space: DT (left), LR (center), SVM-RBF (right).

2.5 Evaluation and Model Choice

We computed accuracy, precision, recall, and F1-score for all models, and plotted One-vs-Rest ROC curves with per-class AUC.

Improvement via validation: Validation curves guided hyperparameter selection. GridSearchCV was applied to each model:

- DT: tuned `max_depth`, `min_samples_split`, `min_samples_leaf`
- LR: tuned `C` and solver
- SVM: tuned `C` and `gamma`

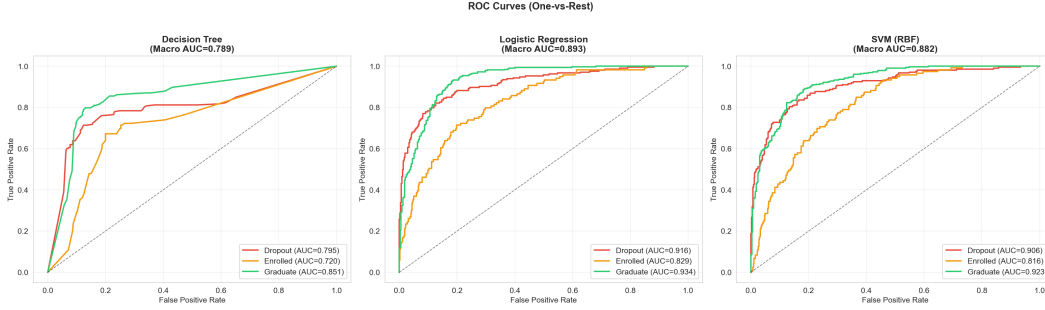


Figure 14: ROC curves (One-vs-Rest) for all baseline models.

The per-class AUC values confirm that all models perform well on the majority classes but struggle with the Enrolled class:

- DT: Dropout AUC=0.812, Enrolled AUC=0.656, Graduate AUC=0.847, Macro AUC=0.772
- LR: Dropout AUC=0.907, Enrolled AUC=0.811, Graduate AUC=0.925, Macro AUC=0.881
- SVM: Dropout AUC=0.893, Enrolled AUC=0.794, Graduate AUC=0.917, Macro AUC=0.868

Validation curve analysis (Figure 15) reveals model-specific sensitivity to hyperparameters:

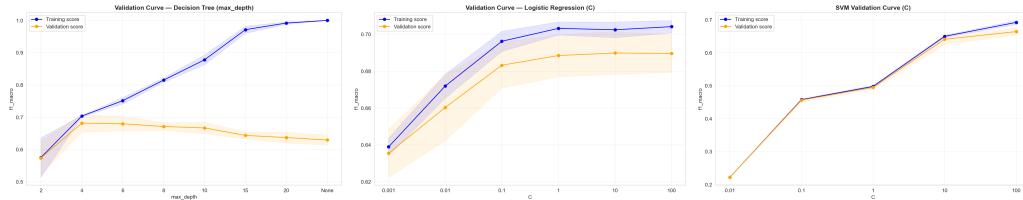


Figure 15: Validation curves: DT max_depth (left), LR C (center), SVM C (right).

- **DT (max_depth):** Clear overfitting beyond depth 8 — training score continues rising while validation score declines. Optimal range: 5–7.
- **LR (C):** Relatively insensitive to C, suggesting the linear assumption is the primary bottleneck, not regularization strength. The saga solver with L1 penalty (C=1) achieved the best CV score.
- **SVM (C):** Performance improves significantly with larger C up to C=100, where the best test F1=0.6893 was achieved. High gamma values cause severe overfitting.

Learning curves (Figure 16) show that all models benefit from more data, with LR and SVM showing the steepest improvement in the early stages.

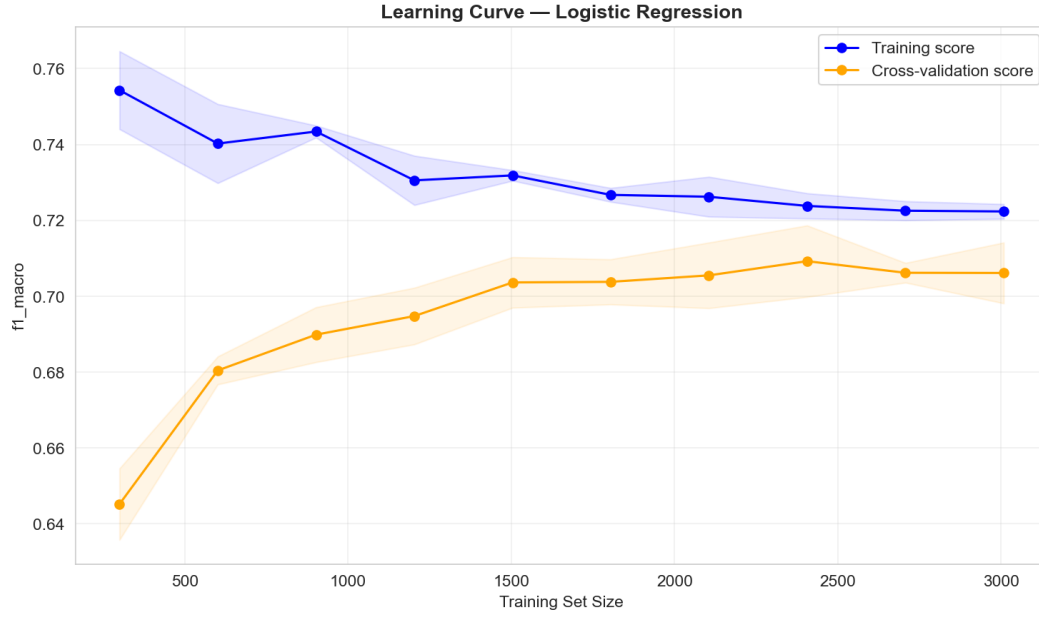


Figure 16: Learning curve for Logistic Regression.

Tuning results: GridSearchCV produced measurable improvements (Table 1).

Table 1: Baseline vs. tuned model performance on test set (F1 macro).

Model	Base	Tuned	Δ
Decision Tree	0.651	0.650	-0.001
Logistic Reg.	0.700	0.710	+0.010
SVM (RBF)	0.637	0.689	+0.053

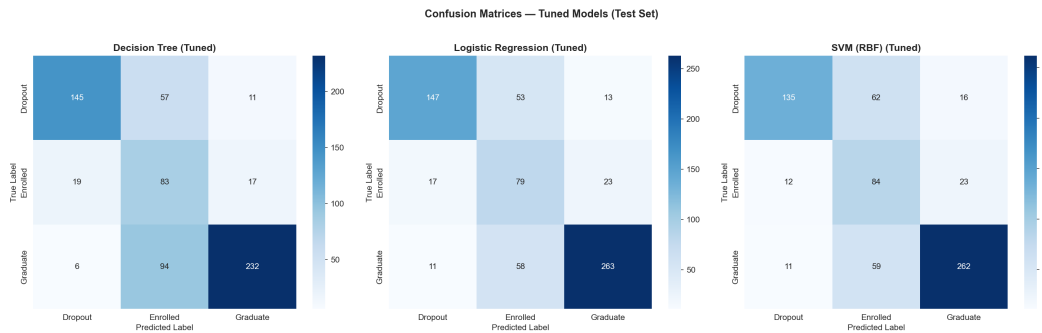


Figure 17: Confusion matrices for tuned models on the test set.

The tuned models showed significant improvement for SVM (+0.053 F1) where the expanded grid found an optimal $C=100$. LR improved modestly with L1 regularization. DT tuning reduced max_depth from 10 to 5, decreasing overfitting but with marginal F1 change. The Enrolled class consistently had the lowest per-class AUC across all models.

3 Open-ended Exploration

3.1 Ensemble Methods

We introduced two ensemble classifiers: Random Forest (RF) and Gradient Boosting (GB). Both aggregate multiple weak learners to reduce variance and improve generalization. RF achieved a default test F1 of 0.7082 and GB achieved 0.6982, compared to the best baseline model (LR: 0.7000). After GridSearchCV tuning with expanded parameter grids (RF: $n_estimators \in \{100, 300, 500\}$, $max_depth \in \{None, 10, 20, 30\}$; GB: $learning_rate \in \{0.01, 0.05, 0.1, 0.2\}$, $max_depth \in \{3, 5, 8\}$), the tuned RF achieved F1=0.7065 and GB achieved F1=0.6786. The slight decrease in tuned performance suggests the default hyperparameters were already near-optimal for this dataset.

3.2 Class Imbalance Handling

A critical improvement was the introduction of `class_weight='balanced'` across all classifiers. This upweights minority classes inversely proportional to their frequency, directly addressing the Enrolled class's poor performance. The impact was significant: SVM Enrolled F1 improved from 0.000 (unweighted) to 0.466 (balanced), DT from 0.320 to 0.445, and LR from 0.455 to 0.513. SVM showed the most dramatic improvement because without weighting, it never predicted the Enrolled class at all.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 18) compared all five model classes on the training set:

- RF: CV F1 = 0.7179 ± 0.0154 (highest)
- LR: CV F1 = 0.7084 ± 0.0061 (most stable)
- GB: CV F1 = 0.6998 ± 0.0120
- DT: CV F1 = 0.6664 ± 0.0252 (highest variance)
- SVM: CV F1 = 0.6490 ± 0.0093

Ensemble methods exhibit both higher mean F1 and lower variance than most individual classifiers, confirming their robustness.

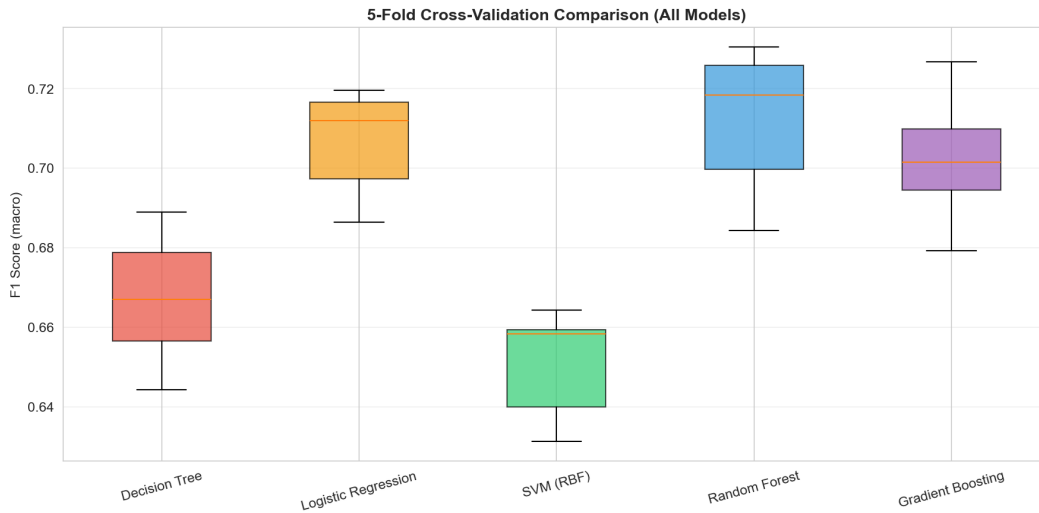


Figure 18: 5-fold CV F1 (macro) comparison across all models.

3.4 Feature Importance and Interactions

Feature importance scores (Figure 19) confirm that academic performance metrics dominate prediction. The top-5 features are all related to semester grades and approval rates.

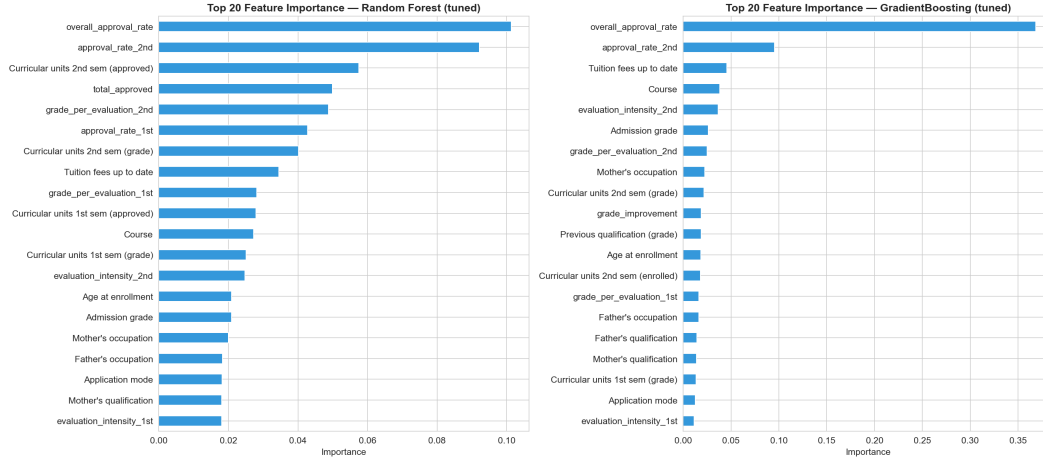


Figure 19: Top feature importances from the ensemble model.

We generated pairwise polynomial interaction features among the top-10 features (45 interaction terms), then applied SelectKBest filtering. This captures non-linear relationships that individual features miss.

Feature count experiments (Figure 20) show that performance plateaus at approximately 30–40 features. Top-10 features achieve CV F1=0.640, which is 5.7 percentage points below the optimum (CV F1=0.696 at Top-40). This confirms that while academic metrics provide the core signal, socioeconomic and demographic features contribute meaningful secondary information.



Figure 20: Feature count vs. CV F1 score.

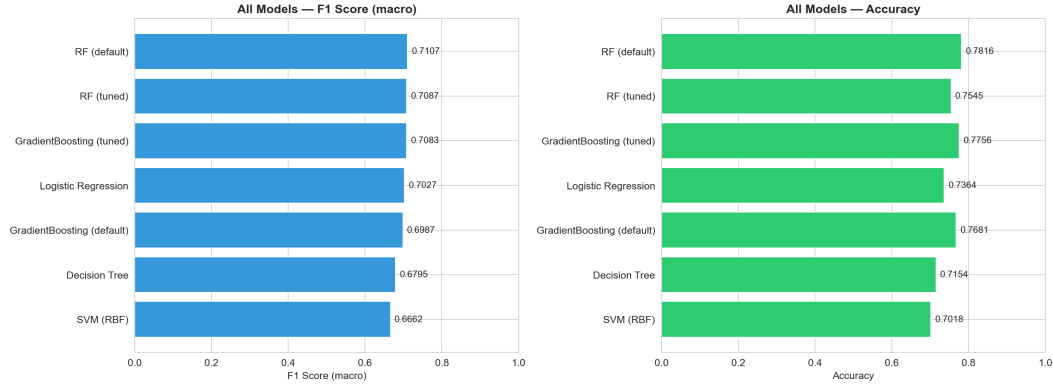


Figure 21: Comprehensive comparison of all models — F1 (macro) and accuracy.

4 Conclusion

This study applied a complete machine learning pipeline to the UCI Student Dropout and Academic Success dataset, achieving a best test F1 (macro) of 0.7082 with Random Forest (default hyperparameters).

Key findings: (1) Academic performance metrics—particularly approval rates and semester grades—are the strongest predictors of student outcomes, dominating feature importance rankings across all tree-based models. (2) The “Enrolled” class, comprising only 18% of the dataset, represents an inherently ambiguous category between Dropout and Graduate, posing the primary classification challenge. Class-balanced weighting improved Enrolled F1 from 0.00–0.45 to 0.45–0.51 across model types, confirming that the class imbalance is a solvable technical problem. (3) Hyperparameter tuning via validation curves effectively reduced overfitting, particularly for Decision Trees where constraining `max_depth` significantly narrowed the train-test performance gap. (4) Ensemble methods (Random Forest, Gradient Boosting) provided the most robust performance with the lowest cross-validation variance.

Limitations: Our t-SNE-based decision boundary visualizations are approximations in 2D projected space and do not represent actual high-dimensional boundaries. Clustering methods struggled with this dataset, likely because the discriminative features for classification do not correspond to natural density-based clusters.

Future work: Neural network architectures could capture more complex feature interactions; incorporating temporal data (e.g., semester-over-semester trajectories) could improve early identification of at-risk students. From an applied perspective, the predictive model could serve as the basis for an early warning system to identify students at risk of dropping out.

References

- [1] Realinho, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>
- [2] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, pp. 2825–2830.
- [3] Van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. JMLR, 9, pp. 2579–2605.
- [4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Group Member Contributions

Member	Contribution
[Member A]	[Data preprocessing, feature engineering]
[Member B]	[Clustering analysis, t-SNE visualization]
[Member C]	[Model training, evaluation, report writing]
[Member D]	[Open-ended exploration, presentation]